Data Schemas

The app reads two configuration files that ship inside it, and keeps one database on the phone. The machine-readable schemas are the source of truth:

Data	File in the app	Machine-readable schema	Checked by
Drug catalog (which drugs are covered)	`data/drug_catalog.json`	`data/schemas/drug_catalog.schema.json` (JSON Schema draft-07)	`mobile/src/rules/catalog.ts`, `npm run config:validate`
Exclusion policy (which drugs are never checked)	`data/exclusion_policy.json`	`data/schemas/exclusion_policy.schema.json`	same
On-phone database (saved checks)	encrypted SQLite on the phone	`mobile/src/storage/migrations.ts`	storage tests
Extended-list review table (proposal only)	`data/proposals/extended_drug_list_review.csv`	this page, section 4	`scripts/catalog/`

Beyond JSON Schema, the validator also checks references (every drug's ingredients exist), duplicate names and codes, catalog × policy overlaps, approval status and version compatibility, and runs a precedence self-test: every active drug must be reachable, every exclusion must win.

1. Drug catalog (schema 2)

File header

Field	Type	Required	Meaning
`schema_version`	`2`	yes	Format version; an app that does not know it refuses the file
`catalog_version`	version string, e.g. `0.3.0-proposed`	yes	Recorded with every saved result
`status`	`APPROVED` · `DRAFT_PENDING_APPROVAL` · `PROPOSED_PENDING_APPROVAL` · `RETIRED`	yes	Pilot builds use `APPROVED` only
`effective_date`	date or `null`	yes	Set on approval
`compatibility.min_ruleset_version`	version	yes	Oldest app rules that can read the file
`description`, `provenance`	text, object	no	Where the scope and codes came from
`scope_defaults`	object	yes	Defaults for every drug: `routes` (`["ORAL"]`), `rx_scope` (`PRESCRIPTION_ONLY`), `ingredient_set_match` (`EXACT`)
`ingredients`	list of Ingredient	yes	Ingredient definitions (below)
`drugs`	list of Drug	yes	What the app covers (below)

Field	Type	Required	Meaning
`change_log`	list of { `date`, `version`, `change`, `source` }	yes	Every change, dated and sourced

Ingredient

Field	Type	Required	Meaning	Example
`id`	lowercase id	yes	Internal key	`clopidogrel`
`name`	text	yes	RxNorm ingredient name	`clopidogrel`
`rxcui`	digits or `null`	yes	RxNorm ingredient code (IN or MIN)	`32968`
`tty`	`IN` · `MIN`	no	RxNorm term type	`IN`
`aliases`	list of text	yes	Other names seen in label or recall text, each with evidence	`["apap"]` for acetaminophen
`alias_provenance`	name → text	no	Where each alias was observed	
`precise_ingredients`	list of { `rxcui`, `name` }	no	Every RxNorm salt form (PIN)	`{"rxcui": "236991", "name": "clopidogrel bisulfate"}`
`provenance`	{ `source`, `verified_on` }	yes	RxNav lookups and date	

Drug

Field	Type	Required	Meaning	Example
`id`	lowercase id	yes	Internal key, shown in audits	`losartan_hydrochlorothiazide`
`display_name`	text	yes	Name shown to people	`Losartan/hydrochlorothiazide`
`status`	`ACTIVE` · `INACTIVE`	yes	`INACTIVE` = known but not checked ("not covered")	`ACTIVE`
`ingredients`	list of ingredient ids	yes	The exact ingredient set; a product with more or fewer ingredients is a different drug	`["losartan", "hydrochlorothiazide"]`
`aliases`	list of text	no	Other drug-level names	
`rx_scope`	`PRESCRIPTION_ONLY` · `OTC_ONLY` · `PRESCRIPTION_AND_OTC`	no	Defaults to the header	`PRESCRIPTION_ONLY`
`restrictions.routes`	list	no	Allowed routes	`["ORAL"]`
`restrictions.dose_forms`	`TABLET` · `CAPSULE` · `SOLUTION` · `SUSPENSION` · `OTHER`	no	Allowed forms	

Field	Type	Required	Meaning	Example
`restrictions.release_types`	`IR` · `ER` · `DR`	no	Allowed release types	`["ER"]` (metoprolol succinate)
`restrictions.strengths`	ingredient → list	no	Allowed strengths	
`restrictions.precise_ingredients`	ingredient → {`allow` or `deny`: RxCUIs}	no	Salt limits	`{"metoprolol": {"allow": ["221124"]}}`
`part_d_rank_2023`	number	no	CMS Part D 2023 rank by total claims	`81`
`provenance`	{`source`, `verified_on`}	yes	Ranking, RxNav and FDA sources and date	
`notes`	text	no	Scope notes, CMS names, product counts, why inactive	

Two drugs with the same ingredient set must have disjoint `dose_forms` or `release_types` (the validator enforces it), so a product can never match two drugs.

2. Exclusion policy (schema 2)

Part	Fields	Meaning
Header	`schema_version`, `policy_version`, `status`, `effective_date`, `last_reviewed`, `next_review_due`, `review_interval_months`, `compatibility`, `source_document`, `principles`	Versioned, dated, reviewed every 6 months
`classes[]`	`id`, `criterion`, `rationale`, `explanation` (shown to users), `fda_epc_signals` (FDA pharmacologic classes), `unlisted_signal_behavior` (`NEEDS_CLARIFICATION` · `EXCLUDED` · `IGNORE`)	Exclusion by published class, e.g. `ANTICOAGULANT`
`ingredient_rules[]`	`id`, `ingredient`, `rxcuis`, `names`, `class`, `decision` (`EXCLUDED` · `NEEDS_CLARIFICATION`), `applies_to` (routes, dose forms, release types), `include_combinations`, `source`, `verified_on`, `note`	One rule per excluded ingredient, e.g. `EXCL-WARFARIN` (RxCUI 11289)
`route_rules[]`	`id`, `status` (`OPEN_DECISION` · `APPROVED`), `decision_id`, `routes`, `scope`, `decision`, `interim_decision`, `interim_reason`, `statement`	The administration rule (OD-1), interim "not covered"
`product_type_rules[]`	`id`, `product_types`, `decision`, `explanation`, `source`, `decided_on`	e.g. `OTC-MVP`: over-the-counter products excluded
`change_log[]`	`date`, `version`, `change`, `source`, `approved_by`	Every change, dated and sourced

Order of evaluation: data valid → ingredient rules → route rules → product type → class signals → catalog. Exclusions always win over the catalog.

3. On-phone database (SQLite, encrypted; schema version 1)

Off in development builds until encryption is verified on a device. Never stores photos, label text, patient names, pharmacy details, Rx numbers or device identifiers.

Table	Columns	Holds
`schema_meta`	`key` (PK), `value`	Database schema version
`medication`	`id` (PK), `display_name`, `catalog_drug_id`, `identity_key`, `created_at`, `archived_at`	One row per medication the user keeps
`fill`	`id` (PK), `medication_id` → medication, `scanned_at`, `fill_date`, `package_ndc`, `product_ndc`, `identity_level` (`PRODUCT` · `CLINICAL_DRUG`), `identity_json`, `dose_form`, `release_mechanism` (`IR` · `ER` · `DR` · `UNSPECIFIED`), `labeler_name`, `lot`, `container_type` (`PHARMACY_VIAL` · `ORIGINAL_PACKAGE` · `UNKNOWN`)	One row per scanned bottle
`assessment`	`id` (PK), `fill_id` → fill, `evaluated_at`, `saved_at`, `finding` (`CONFIRMED_MATCH` · `POTENTIAL_MATCH` · `NO_MATCH_FOUND` · `UNRESOLVED`), `finding_json`, `ruleset_version`, `policy_version`, `catalog_version`, `dataset_last_updated`, `coverage_complete` (0/1)	One row per saved recall check
`assessment_evidence`	`assessment_id` → assessment, `recall_number` (PK together), `relevance` (`PRODUCT_MATCH` · `POSSIBLE_MATCH` · `OTHER_PRODUCT` · `UNPARSEABLE`), `classification`, `lot_result`, `reasons`	Each recall record behind a check
`source_cache`	`source`, `query_key` (PK together), `outcome` (`OK` · `NO_MATCH`), `body_json`, `source_last_updated`, `retrieved_at`, `expires_at`	Short-lived cache of public FDA/NIH answers; never errors
`config_snapshot`	`kind` (`CATALOG` · `POLICY`), `version` (PK together), `json`, `installed_at`	Each catalog and policy version results were made under

Deleting a medication deletes its fills, checks and evidence. "Delete all data on this phone" erases the database and its key.

4. Extended-list review table (proposal)

`data/proposals/extended_drug_list_review.csv`, one row per drug (CMS names of the same ingredient merged):

Column	Meaning
`rank`	CMS Part D 2023 rank by total claims (first name in the row)
`cms_generic_names`	CMS generic name(s) as published
`claims_2023`, `share_pct`	Total claims, and share of all 1,616,932,462 Part D claims in 2023
`ingredients_rxcui`	RxNorm ingredient(s) and RxCUI, verified exactly on RxNav
`decision`	In the app now · Add (active) · Add (switched off) · Excluded by policy · Out of scope
`catalog_drug_id`	Drug `id` in `data/proposals/drug_catalog_0.3.0-proposed.json`
`fda_oral_rx_products`	Oral prescription products with exactly this ingredient set in FDA's NDC Directory

Column	Meaning
`recall_records_app_search`	Recall records the app's recall search finds today (openFDA enforcement)
`example_ndc`	The real FDA product used to check the drug with the app's own code
`reason`	Why a drug is excluded, out of scope or switched off